Name:_____ Period:_____
Earth and Space Science

GRAVITY, ORBITS, AND PLANETARY MOTION LEARNING GUIDE

Learning Intention
Today we will be learning how gravity and inertia keep objects in orbit and how Kepler's laws describe the shape, speed, and timing of orbits so we can predict how planets, moons, and spacecraft move through the solar system.

Success Criteria
I can… ● explain how mass and distance affect gravity and why an orbiting object follows a curved path. ● calculate the eccentricity of an orbit and identify where the Sun is located in a planet's orbit. ● use Kepler's laws to predict how an object's speed and period of revolution change with its distance from the object it orbits.

Artemis II: Falling Around the Moon

On April 1, 2026, NASA launched four astronauts, Reid Wiseman, Victor Glover, Christina Koch, and Jeremy Hansen, toward the Moon aboard the Orion spacecraft. They never landed. Instead, Orion swung around the far side of the Moon and headed home, reaching 252,756 miles from Earth, farther than any humans had traveled before. For most of the trip, Orion simply coasted. The crew floated inside the cabin, but they were never in "zero gravity." The gravity of Earth and the Moon shaped every mile of their path. *How can gravity keep a spacecraft, a moon, or a planet moving along a curved path without an engine pushing it?*

Gravity: A Pull Between Every Object

Gravity is a force of attraction between all objects that have mass. Every object pulls on every other object. The strength of that pull depends on two things:

- **Mass:** The more mass the objects have, the stronger the pull. The Sun contains more than 99% of all the mass in the solar system, so its gravity controls the motion of every planet.
- **Distance:** The closer the objects are, the stronger the pull. Gravity gets weaker quickly as distance increases. If the distance between two objects doubles, the pull becomes only one-fourth as strong. Distance can matter more than mass. Earth has about 81 times more mass than the Moon. But as Orion got close to the Moon, the Moon's gravity pulled on the spacecraft more strongly than Earth's gravity did, because Orion was now much closer to the Moon than to Earth.

Check for Understanding

_____ 1. Which change would increase the gravitational force between two objects?
(1) moving the objects farther apart
(2) decreasing the mass of both objects
(3) moving the objects closer together
(4) decreasing the mass of one object

_____ 2. The Sun's gravity controls the motion of every planet in the solar system mainly because the Sun
(1) is the hottest object in the solar system
(2) has by far the most mass in the solar system
(3) is the closest object to every planet
(4) gives off light and heat

3. Earth has about 81 times more mass than the Moon. Explain how the Moon's gravity could pull on Orion more strongly than Earth's gravity did.

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What Keeps an Object in Orbit?

A moving object keeps moving in a straight line at the same speed unless a force acts on it. This is called **inertia**. Space has almost no air to slow things down, so a moving object would travel in a straight line forever if nothing pulled on it.

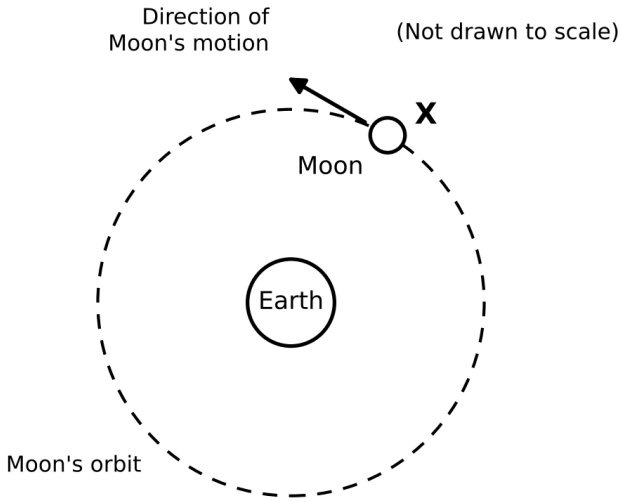
An **orbit** is the curved path of one object around another. An orbit forms when inertia and gravity work together. Inertia carries the object forward, and gravity pulls it toward the object it orbits, bending the straight path into a curve. An orbiting object is actually falling the whole time, but it moves sideways so fast that it keeps missing the object it is falling toward. The Moon has been falling toward Earth for billions of years.

Astronauts float for the same reason. The crew and Orion were falling together, so nothing pressed the astronauts against the floor, even though gravity was pulling on them the whole time. If gravity suddenly disappeared, an orbiting object would fly off in a straight line in the direction it was moving at that moment. If the object stopped moving sideways, gravity would pull it straight in.

Any object that orbits another object is called a **satellite**. The Moon is a natural satellite of Earth, and the International Space Station is a human-made satellite. The same rules of gravity apply to both. Spacecraft sent to other planets are already moving with Earth around the Sun, so they travel along curved paths and are aimed at where the planet will be when they arrive, not where it is on launch day.

Check for Understanding: Model It

The diagram shows the Moon at position X in its orbit around Earth. The arrow shows the direction the Moon is moving.



1. a. **Draw an arrow** starting at X to show the direction of Earth's gravitational pull on the Moon. **Label** it **gravity**.
   b. **Draw a dashed line** starting at X to show the path the Moon would take if Earth's gravity suddenly disappeared.

_____ 2. Astronauts aboard Orion floated inside the cabin because
(1) there is no gravity in space
(2) the Moon's gravity canceled out Earth's gravity
(3) the astronauts and the spacecraft were falling together
(4) Orion's engines pushed them upward

Kepler's First Law: Orbits Are Ellipses

In the early 1600s, Johannes Kepler used Tycho Brahe's careful observations of Mars to discover three laws that describe how planets move. Scientists still use these laws to plan missions today. Kepler's first law states that every planet orbits the Sun in an ellipse, with the Sun at one focus.

- **Ellipse:** an oval-shaped closed curve. A circle is a special kind of ellipse.
- **Foci** (singular: **focus**): two fixed points inside an ellipse. The Sun is at one focus of each planet's orbit. The other focus is empty space.
- **Major axis:** the longest line across an ellipse. It passes through both foci.
- **Perihelion:** the point in an orbit where the planet is closest to the Sun.
- **Aphelion:** the point in an orbit where the planet is farthest from the Sun.

Eccentricity describes how oval, or elongated, an orbit is. It has no units.

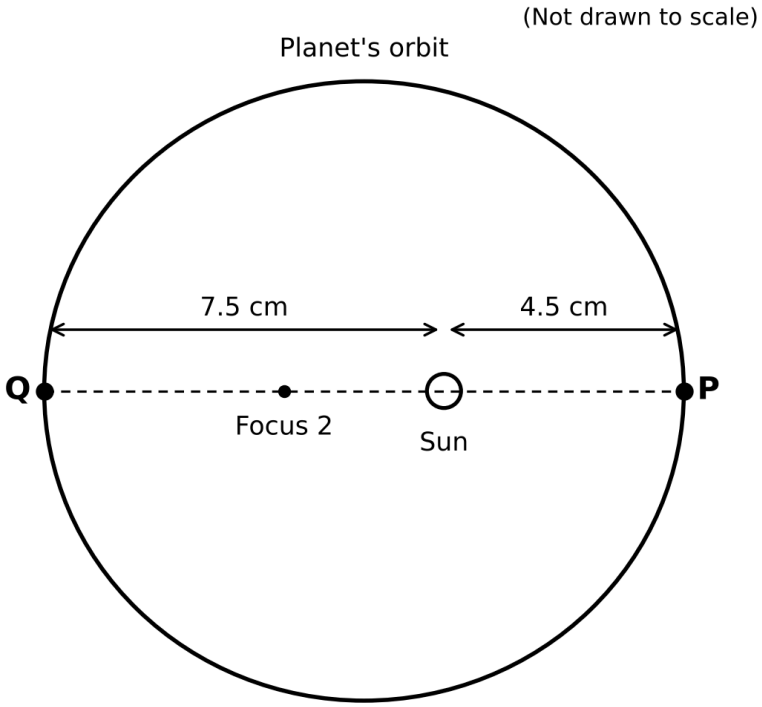
Eccentricity = distance between foci ÷ length of major axis

Eccentricity is always between 0 and 1. A circle has an eccentricity of 0 because its two foci are at the same point. As the foci move farther apart, the ellipse gets flatter and its eccentricity gets closer to 1. Most planets have orbits that are very close to circles. Earth's eccentricity is only 0.017.

You can rearrange the formula to find a missing value: distance between foci = eccentricity × length of major axis. If you know the distances from the Sun to the aphelion and to the perihelion, subtract them to find the distance between the foci. Add them to find the length of the major axis.

Check for Understanding

The diagram below is a model of a planet's orbit. The Sun is at one focus. Points P and Q are at the ends of the major axis.



1. Which letter shows the planet's perihelion?

   _____

2. Calculate the length of the major axis.

   _____ cm

3. Calculate the distance between the foci.

_____ cm

4. Calculate the eccentricity of this orbit. Round to the nearest thousandth.

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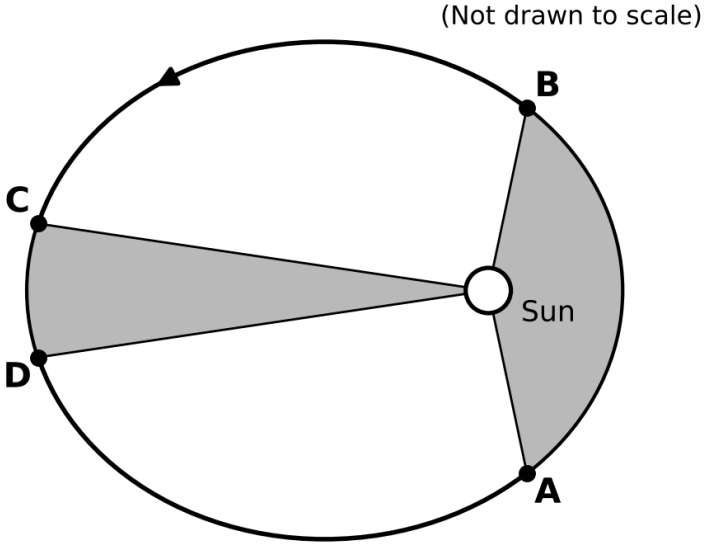
_____ 5. If the foci in this model were moved closer together, the orbit would become
(1) more elongated, and its eccentricity would increase
(2) more elongated, and its eccentricity would decrease
(3) more circular, and its eccentricity would increase
(4) more circular, and its eccentricity would decrease

6. A model of Mars's orbit has a major axis that is 20.0 cm long. The eccentricity of Mars's orbit is 0.094. Calculate the distance between the foci in this model. Round to the nearest tenth.

_____ cm

Kepler's Second Law: Faster When Closer

Kepler's second law states that a line connecting a planet to the Sun sweeps out equal areas in equal amounts of time. In the diagram below, the two shaded areas are equal, so the planet takes the same amount of time to travel from A to B as it does from C to D.



The distance from A to B is much longer than the distance from C to D, so the planet must move faster near perihelion and slower near aphelion. Gravity explains why. As a planet moves toward the Sun, the Sun's stronger pull speeds it up. As it moves away, the Sun's pull slows it down.

The Parker Solar Probe shows this law in action. Its orbit is a very eccentric ellipse. On December 24, 2024, it passed about 3.8 million miles above the Sun's surface at roughly 430,000 miles per hour, faster than any other human-made object has ever moved. Far from the Sun, the same probe moves much more slowly, even though no engine changes its speed.

Check for Understanding

_____ 1. At which point in its orbit does a planet move fastest?
(1) aphelion
(2) perihelion
(3) halfway between perihelion and aphelion
(4) The speed is the same at every point.

2. Explain why the Sun's gravitational pull on a planet is stronger at perihelion than at aphelion.

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Kepler's Third Law: Farther Means Longer

The **period of revolution** is the time it takes an object to complete one orbit. A planet's period of revolution is the length of its year. Kepler's third law states that the farther a planet is from the Sun, the longer its period of revolution. Planets farther from the Sun have longer paths to travel, and they move more slowly because the Sun's gravity is weaker far away. This is true for any satellite: the closer it is to the object it orbits, the faster it moves.

Planet	Mean Distance from Sun (million km)	Period of Revolution (Earth years)
Mercury	58	0.24
Earth	150	1.00
Mars	228	1.88
Jupiter	779	11.86
Neptune	4,495	164.8

Data: NASA, values rounded

Kepler found an exact pattern. Measure distance in astronomical units (AU), where 1 AU is Earth's average distance from the Sun, and measure period in Earth years. Then $(\text{period})^2 = (\text{distance})^3$. For example, an object 9 AU from the Sun has a period of 27 years, because $27^2 = 729$ and $9^3 = 729$.

Check for Understanding

_____ 1. Based on the data table, as a planet's distance from the Sun increases, its period of revolution
(1) decreases
(2) increases
(3) stays the same
(4) increases, then decreases

2. An asteroid orbits the Sun at an average distance of 4 AU. Use Kepler's third law to calculate its period of revolution.

_____ Earth years

ESSRT Connection

Use the **Solar System Objects Data Table** (page 2) in the 2026 Revised Edition Reference Tables for Earth and Space Sciences to answer questions 1 through 3.

1. Which planet has the most eccentric (least circular) orbit?

_____

_____ 2. Which planet's orbit is most nearly a perfect circle?
(1) Mercury
(2) Venus
(3) Mars
(4) Neptune

3. Compare the mean distance from the Sun and the period of revolution of Mars and Jupiter. Use Kepler's third law to explain the difference.

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Cluster Set 1

Base your answers to questions 1 through 4 on the data table below and on your knowledge of Earth and Space Science.

The data table shows the average distance from Earth's center, the period of revolution, and the average orbital speed of four satellites of Earth.

Satellite	Distance from Earth's Center (km)	Period of Revolution	Average Orbital Speed (km/h)
International Space Station	6,780	1.5 hours	27,600
GPS satellite	26,600	12 hours	13,900
GOES weather satellite	42,200	24 hours	11,100
Moon	384,400	27.3 days	3,680

Data: NASA and NOAA, values rounded

_____ 1. Which statement best describes the pattern in the data?
(1) As distance increases, period decreases and speed increases.
(2) As distance increases, period increases and speed decreases.
(3) As distance increases, period and speed both increase.
(4) As distance increases, period and speed both decrease.

2. Calculate how many times the International Space Station orbits Earth in one day (24 hours).

_____ orbits

_____ 3. A new satellite is placed in orbit 15,000 km from Earth's center. Its period of revolution would most likely be
(1) less than 1.5 hours
(2) between 1.5 hours and 12 hours
(3) between 12 hours and 24 hours
(4) more than 27.3 days

4. Use your knowledge of gravity to explain why the International Space Station travels much faster than the Moon.

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Mini Glossary

Aphelion – the point in an orbit where an object is farthest from the Sun
Eccentricity – a number from 0 to 1 that describes how oval an orbit is; distance between foci ÷ length of major axis
Ellipse – an oval-shaped closed curve with two foci
Focus (plural: foci) – one of two fixed points inside an ellipse; the Sun is at one focus of each planet's orbit
Gravity – a force of attraction between all objects that have mass; stronger with more mass and less distance
Inertia – the tendency of a moving object to keep moving in a straight line at the same speed unless a force acts on it
Major axis – the longest line across an ellipse, passing through both foci
Orbit – the curved path of one object around another, caused by gravity and inertia working together
Perihelion – the point in an orbit where an object is closest to the Sun
Period of revolution – the time it takes an object to complete one orbit
Satellite – any object that orbits another object, natural or human-made